

Complete Course: Linear Algebra and Functional Analysis

Detailed Reference Guide

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1 Introduction to Linear Algebra

Linear algebra is the mathematical foundation for data science, machine learning, signal processing, and many other fields. It studies vector spaces and linear mappings between them.

1.1 Historical Context

Linear algebra emerged from the study of systems of linear equations. The term "matrix" was coined by James Joseph Sylvester in 1850, and much of the modern notation was developed by Arthur Cayley.

2 Vector Spaces

2.1 Formal Definition

Definition 2.1 (Vector Space). A **vector space** V over a field $\mathbb{F}$ (typically $\mathbb{R}$ or $\mathbb{C}$) is a set equipped with two operations:

1. **Vector addition:** $+: V \times V \rightarrow V$
2. **Scalar multiplication:** $\cdot: \mathbb{F} \times V \rightarrow V$

satisfying the following axioms for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all $\alpha, \beta \in \mathbb{F}$:

1. **Associativity of addition:** $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
2. **Commutativity of addition:** $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. **Identity element of addition:** $\exists \mathbf{0} \in V$ such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$
4. **Inverse elements of addition:** $\forall \mathbf{v} \in V, \exists -\mathbf{v} \in V$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$
5. **Compatibility of scalar multiplication:** $\alpha(\beta\mathbf{v}) = (\alpha\beta)\mathbf{v}$
6. **Identity element of scalar multiplication:** $1\mathbf{v} = \mathbf{v}$
7. **Distributivity of scalar multiplication w.r.t vector addition:** $\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$
8. **Distributivity of scalar multiplication w.r.t field addition:** $(\alpha + \beta)\mathbf{v} = \alpha\mathbf{v} + \beta\mathbf{v}$

2.2 Examples of Vector Spaces

Example 2.2 (Common Vector Spaces). 1. $\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{R}\}$ with component-wise addition and scalar multiplication

2. $\mathbb{C}^n = \{(z_1, z_2, \dots, z_n) : z_i \in \mathbb{C}\}$

3. $M_{m \times n}(\mathbb{R})$: all $m \times n$ matrices with real entries
4. $\mathcal{P}_n$: polynomials of degree $\leq n$ with real coefficients
5. $C[a, b]$: continuous functions on interval $[a, b]$

2.3 Subspaces

Definition 2.3 (Subspace). A subset W of a vector space V is a **subspace** if:

1. $\mathbf{0} \in W$
2. Closed under addition: $\mathbf{u}, \mathbf{v} \in W \Rightarrow \mathbf{u} + \mathbf{v} \in W$
3. Closed under scalar multiplication: $\alpha \in \mathbb{F}, \mathbf{v} \in W \Rightarrow \alpha \mathbf{v} \in W$

3 Linear Independence, Basis, and Dimension

3.1 Linear Combination

Definition 3.1 (Linear Combination). Given vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in V$ and scalars $\alpha_1, \alpha_2, \dots, \alpha_k \in \mathbb{F}$, the vector $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_k \mathbf{v}_k$ is called a **linear combination** of the vectors.

3.2 Span

Definition 3.2 (Span). The **span** of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ is the set of all their linear combinations:

$$\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\} = \left\{ \sum_{i=1}^k \alpha_i \mathbf{v}_i : \alpha_i \in \mathbb{F} \right\}$$

3.3 Linear Independence

Definition 3.3 (Linear Independence). Vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are **linearly independent** if the equation

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_k \mathbf{v}_k = \mathbf{0}$$

has only the trivial solution $\alpha_1 = \alpha_2 = \dots = \alpha_k = 0$.

Theorem 3.4 (Test for Linear Independence). In $\mathbb{R}^n$, vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are linearly independent if and only if the matrix $A = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_k]$ has rank k .

Proof. The equation $\sum_{i=1}^k \alpha_i \mathbf{v}_i = \mathbf{0}$ can be written as $A\boldsymbol{\alpha} = \mathbf{0}$, where $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_k)^\top$. This homogeneous system has only the trivial solution if and only if $\text{rank}(A) = k$. $\square$

3.4 Basis

Definition 3.5 (Basis). A set $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a **basis** for vector space V if:

1. B is linearly independent
2. B spans V : $\text{Span}(B) = V$

Theorem 3.6 (Unique Representation). If $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for V , then every vector $\mathbf{v} \in V$ can be written uniquely as:

$$\mathbf{v} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n$$

The coefficients $(\alpha_1, \alpha_2, \dots, \alpha_n)$ are called the **coordinates** of $\mathbf{v}$ relative to basis B .

3.5 Dimension

Definition 3.7 (Dimension). The **dimension** of a vector space V , denoted $\dim(V)$, is the number of vectors in any basis for V .

Theorem 3.8 (Fundamental Theorem of Dimension). All bases for a finite-dimensional vector space have the same number of elements.

Proof. Suppose $B_1 = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ and $B_2 = \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$ are two bases. Since B_1 spans V and B_2 is linearly independent, we must have $m \leq n$. Similarly, since B_2 spans and B_1 is linearly independent, $n \leq m$. Thus $m = n$. $\square$

4 Matrices and Linear Transformations

4.1 Matrix as a Linear Transformation

Definition 4.1 (Linear Transformation). A function $T : V \rightarrow W$ between vector spaces is **linear** if:

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in V$
2. $T(\alpha \mathbf{v}) = \alpha T(\mathbf{v})$ for all $\alpha \in \mathbb{F}, \mathbf{v} \in V$

Theorem 4.2 (Matrix Representation). Every linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be represented by an $m \times n$ matrix A such that $T(\mathbf{x}) = A\mathbf{x}$.

Proof. Let $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ be the standard basis for $\mathbb{R}^n$. Define A such that its j -th column is $T(\mathbf{e}_j)$. Then for any $\mathbf{x} = \sum_{j=1}^n x_j \mathbf{e}_j$:

$$T(\mathbf{x}) = T\left(\sum_{j=1}^n x_j \mathbf{e}_j\right) = \sum_{j=1}^n x_j T(\mathbf{e}_j) = \sum_{j=1}^n x_j \mathbf{a}_j = A\mathbf{x}$$

where $\mathbf{a}_j$ is the j -th column of A . $\square$

4.2 Matrix Operations

4.2.1 Matrix Multiplication

For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, the product $C = AB \in \mathbb{R}^{m \times p}$ is defined by:

$$c_{ij} = \sum_{k=1}^n a_{ik}b_{kj} \quad \text{for } i = 1, \dots, m, \quad j = 1, \dots, p$$

Theorem 4.3 (Properties of Matrix Multiplication). *For matrices of compatible dimensions:*

1. *Associative:* $(AB)C = A(BC)$
2. *Distributive:* $A(B + C) = AB + AC$
3. $(AB)^\top = B^\top A^\top$
4. *Generally NOT commutative:* $AB \neq BA$ in general

4.2.2 Matrix Transpose

For $A = [a_{ij}] \in \mathbb{R}^{m \times n}$, the **transpose** $A^\top \in \mathbb{R}^{n \times m}$ is defined by $(A^\top)_{ij} = a_{ji}$.

Theorem 4.4 (Properties of Transpose). 1. $(A^\top)^\top = A$

2. $(A + B)^\top = A^\top + B^\top$
3. $(\alpha A)^\top = \alpha A^\top$
4. $(AB)^\top = B^\top A^\top$
5. *If A is invertible,* $(A^{-1})^\top = (A^\top)^{-1}$

4.3 Rank of a Matrix

Definition 4.5 (Rank). The **rank** of a matrix A , denoted $\text{rank}(A)$, is:

1. *The maximum number of linearly independent rows*
2. *The maximum number of linearly independent columns*
3. *The dimension of the column space (range) of A*
4. *The dimension of the row space of A*

Theorem 4.6 (Rank-Nullity Theorem). *For $A \in \mathbb{R}^{m \times n}$:*

$$\text{rank}(A) + (A) = n$$

where $(A) = \dim(\text{null}(A))$ is the dimension of the null space.

Proof. Let $\text{rank}(A) = r$. Then the row reduced echelon form of A has r pivot columns and $n - r$ free variables, which correspond to the null space dimension. $\square$

Theorem 4.7 (Rank Inequalities). *For matrices of compatible dimensions:*

1. $\text{rank}(A) \leq \min(m, n)$ for $A \in \mathbb{R}^{m \times n}$
2. $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$
3. $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$
4. $\text{rank}(AB) \geq \text{rank}(A) + \text{rank}(B) - n$ (Sylvester's inequality)

4.4 Determinants

Definition 4.8 (Determinant). *The **determinant** of an $n \times n$ matrix $A = [a_{ij}]$ is:*

$$\det(A) = \sum_{\pi \in S_n} \text{sgn}(\pi) \prod_{i=1}^n a_{i, \pi(i)}$$

where:

- S_n is the set of all permutations of $\{1, 2, \dots, n\}$
- $\text{sgn}(\pi)$ is the sign of permutation π (+1 for even, -1 for odd)

4.4.1 Properties of Determinants

Theorem 4.9 (Properties). *For $n \times n$ matrices:*

1. $\det(I_n) = 1$
2. $\det(A^\top) = \det(A)$
3. $\det(AB) = \det(A) \det(B)$
4. $\det(A^{-1}) = \frac{1}{\det(A)}$ if A is invertible
5. $\det(\alpha A) = \alpha^n \det(A)$
6. Adding a multiple of one row to another doesn't change the determinant
7. Swapping two rows multiplies determinant by -1
8. Multiplying a row by α multiplies determinant by α

4.4.2 Computing Determinants

Theorem 4.10 (Laplace Expansion). *For any $i \in \{1, \dots, n\}$:*

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(A_{ij})$$

where A_{ij} is the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j .

4.4.3 Geometric Interpretation

The determinant represents the signed volume scaling factor of the linear transformation. For columns $\mathbf{a}_1, \dots, \mathbf{a}_n$:

$$|\det(A)| = \text{Volume of parallelepiped spanned by } \mathbf{a}_1, \dots, \mathbf{a}_n$$

4.5 Matrix Inverse

Definition 4.11 (Inverse Matrix). *An $n \times n$ matrix A is **invertible** if there exists B such that $AB = BA = I_n$. This B is unique and denoted A^{-1} .*

Theorem 4.12 (Conditions for Invertibility). *For $n \times n$ matrix A , the following are equivalent:*

1. A is invertible
2. $\det(A) \neq 0$
3. $\text{rank}(A) = n$
4. The columns of A are linearly independent
5. The rows of A are linearly independent
6. The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution
7. The equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b} \in \mathbb{R}^n$

4.5.1 Computing the Inverse

Theorem 4.13 (Formula using Adjugate).

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

where $\text{adj}(A)$ is the adjugate matrix with $\text{adj}(A)_{ij} = (-1)^{i+j} \det(A_{ji})$.

4.5.2 Gauss-Jordan Elimination

To find A^{-1} :

1. Form augmented matrix $[A \mid I]$
2. Perform row operations to get $[I \mid B]$
3. Then $B = A^{-1}$

5 Inner Product Spaces

5.1 Definition and Properties

Definition 5.1 (Inner Product). An **inner product** on a real vector space V is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ satisfying:

1. *Symmetry*: $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
2. *Linearity*: $\langle \alpha \mathbf{u} + \beta \mathbf{v}, \mathbf{w} \rangle = \alpha \langle \mathbf{u}, \mathbf{w} \rangle + \beta \langle \mathbf{v}, \mathbf{w} \rangle$
3. *Positive definiteness*: $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$ with equality iff $\mathbf{v} = \mathbf{0}$

For complex spaces, replace symmetry with conjugate symmetry: $\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle}$.

5.2 Euclidean Inner Product

In $\mathbb{R}^n$, the standard inner product is:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i$$

In $\mathbb{C}^n$:

$$\langle \mathbf{z}, \mathbf{w} \rangle = \mathbf{w}^* \mathbf{z} = \sum_{i=1}^n \overline{w_i} z_i$$

5.3 Norm Induced by Inner Product

Definition 5.2 (Norm). The **norm** induced by an inner product is:

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$$

Theorem 5.3 (Properties of Norm). 1. $\|\mathbf{v}\| \geq 0$ with equality iff $\mathbf{v} = \mathbf{0}$

2. $\|\alpha \mathbf{v}\| = |\alpha| \|\mathbf{v}\|$

3. *Triangle inequality*: $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

5.4 Cauchy-Schwarz Inequality

Theorem 5.4 (Cauchy-Schwarz). *For any vectors $\mathbf{u}, \mathbf{v}$ in an inner product space:*

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

with equality iff $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.

Proof. For any scalar t , consider $\|\mathbf{u} + t\mathbf{v}\|^2 \geq 0$. This gives:

$$\|\mathbf{u}\|^2 + 2t\langle \mathbf{u}, \mathbf{v} \rangle + t^2\|\mathbf{v}\|^2 \geq 0$$

This quadratic in t must have non-positive discriminant:

$$(2\langle \mathbf{u}, \mathbf{v} \rangle)^2 - 4\|\mathbf{u}\|^2\|\mathbf{v}\|^2 \leq 0$$

Thus $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$. □

5.5 Orthogonality

Definition 5.5 (Orthogonality). *Vectors $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.*

Theorem 5.6 (Pythagorean Theorem). *If $\mathbf{u} \perp \mathbf{v}$, then:*

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

Proof.

$$\|\mathbf{u} + \mathbf{v}\|^2 = \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$
□

5.6 Orthonormal Basis

Definition 5.7 (Orthonormal Basis). *A basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is **orthonormal** if:*

$$\langle \mathbf{e}_i, \mathbf{e}_j \rangle = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Theorem 5.8 (Coordinates in Orthonormal Basis). *If $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is an orthonormal basis, then any vector $\mathbf{v}$ can be written as:*

$$\mathbf{v} = \sum_{i=1}^n \langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i$$

*The coefficients $\langle \mathbf{v}, \mathbf{e}_i \rangle$ are the **Fourier coefficients**.*

5.7 Orthogonal Matrices

Definition 5.9 (Orthogonal Matrix). A real square matrix Q is **orthogonal** if $Q^\top Q = QQ^\top = I$, i.e., $Q^{-1} = Q^\top$.

Theorem 5.10 (Properties of Orthogonal Matrices). 1. Columns (and rows) form an orthonormal basis

2. Preserves inner products: $\langle Q\mathbf{x}, Q\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$

3. Preserves norms: $\|Q\mathbf{x}\| = \|\mathbf{x}\|$

4. $\det(Q) = \pm 1$

6 Eigenvalues and Eigenvectors

6.1 Basic Definitions

Definition 6.1 (Eigenvalue and Eigenvector). For an $n \times n$ matrix A , a scalar λ is an **eigenvalue** and a non-zero vector $\mathbf{v}$ is the corresponding **eigenvector** if:

$$A\mathbf{v} = \lambda\mathbf{v}$$

Theorem 6.2 (Characteristic Polynomial). λ is an eigenvalue of A if and only if:

$$\det(A - \lambda I) = 0$$

The polynomial $p(\lambda) = \det(A - \lambda I)$ is called the **characteristic polynomial**.

6.2 Algebraic and Geometric Multiplicity

Definition 6.3 (Multiplicities). • **Algebraic multiplicity**: Multiplicity of λ as a root of $p(\lambda)$

• **Geometric multiplicity**: Dimension of eigenspace $E_\lambda = \{\mathbf{v} : A\mathbf{v} = \lambda\mathbf{v}\}$

Theorem 6.4 (Inequality of Multiplicities). For any eigenvalue λ :

$$1 \leq \text{geometric multiplicity} \leq \text{algebraic multiplicity}$$

6.3 Diagonalization

Definition 6.5 (Diagonalizable Matrix). An $n \times n$ matrix A is **diagonalizable** if there exists an invertible matrix P and diagonal matrix D such that:

$$A = PDP^{-1}$$

Theorem 6.6 (Conditions for Diagonalizability). A is diagonalizable if and only if:

1. The sum of geometric multiplicities equals n
2. There exists a basis of $\mathbb{C}^n$ consisting of eigenvectors of A
3. All eigenvalues are semisimple (algebraic = geometric multiplicity)

Proof. If A is diagonalizable with $A = PDP^{-1}$, then columns of P are eigenvectors and diagonal entries of D are eigenvalues. Conversely, if we have n linearly independent eigenvectors, we can form P from them and D from the eigenvalues. $\square$

6.4 Spectral Theorem for Symmetric Matrices

Theorem 6.7 (Spectral Theorem). *For a real symmetric matrix A ($A = A^\top$):*

1. All eigenvalues are real
2. Eigenvectors corresponding to distinct eigenvalues are orthogonal
3. A is orthogonally diagonalizable: $A = QDQ^\top$ where Q is orthogonal

Proof. For (1): If $A\mathbf{v} = \lambda\mathbf{v}$ with $\|\mathbf{v}\| = 1$, then:

$$\lambda = \mathbf{v}^* A \mathbf{v} = \overline{\mathbf{v}^* A \mathbf{v}} = \overline{\lambda}$$

so λ is real. For (3): Use Gram-Schmidt on eigenvectors to get orthonormal basis. $\square$

6.5 Positive Definite Matrices

Definition 6.8 (Positive Definiteness). *A symmetric matrix A is:*

- **Positive definite** if $\mathbf{x}^\top A \mathbf{x} > 0$ for all $\mathbf{x} \neq 0$
- **Positive semidefinite** if $\mathbf{x}^\top A \mathbf{x} \geq 0$ for all $\mathbf{x}$

Theorem 6.9 (Characterizations). *For symmetric A , the following are equivalent:*

1. A is positive definite
2. All eigenvalues are positive
3. All leading principal minors are positive
4. There exists B with linearly independent columns such that $A = B^\top B$

7 Matrix Decompositions

7.1 LU Decomposition

Theorem 7.1 (LU Decomposition). *If A is an $n \times n$ matrix and all leading principal minors are nonzero, then A can be factored as:*

$$A = LU$$

where L is lower triangular with 1's on diagonal, and U is upper triangular.

Algorithm 1 LU Decomposition with Partial Pivoting

```
1: procedure LU-DECOMPOSITION( $A$ )
2:   for  $k = 1$  to  $n - 1$  do
3:     Find  $p \geq k$  such that  $|A_{pk}| = \max_{i \geq k} |A_{ik}|$ 
4:     Swap rows  $k$  and  $p$  of  $A$ 
5:     for  $i = k + 1$  to  $n$  do
6:        $A_{ik} \leftarrow A_{ik}/A_{kk}$ 
7:       for  $j = k + 1$  to  $n$  do
8:          $A_{ij} \leftarrow A_{ij} - A_{ik}A_{kj}$ 
9:       end for
10:    end for
11:  end for
12:  return  $L$  (from lower triangular with 1's on diagonal),  $U$  (upper triangular)
13: end procedure
```

7.2 Cholesky Decomposition

Theorem 7.2 (Cholesky Decomposition). *If A is symmetric positive definite, then there exists a unique lower triangular matrix L with positive diagonal entries such that:*

$$A = LL^\top$$

Proof. By induction. For $n = 1$, $A = [a_{11}]$, take $l_{11} = \sqrt{a_{11}}$. Assume true for $(n-1) \times (n-1)$. Write:

$$A = \begin{bmatrix} A_{n-1} & \mathbf{b} \\ \mathbf{b}^\top & a_{nn} \end{bmatrix}$$

By induction, $A_{n-1} = L_{n-1}L_{n-1}^\top$. Then:

$$A = \begin{bmatrix} L_{n-1} & 0 \\ \mathbf{c}^\top & l_{nn} \end{bmatrix} \begin{bmatrix} L_{n-1}^\top & \mathbf{c} \\ 0 & l_{nn} \end{bmatrix}$$

where $L_{n-1}\mathbf{c} = \mathbf{b}$ and $l_{nn} = \sqrt{a_{nn} - \mathbf{c}^\top \mathbf{c}}$. □

7.3 QR Decomposition

Theorem 7.3 (QR Decomposition). *Every $m \times n$ matrix A with linearly independent columns can be factored as:*

$$A = QR$$

where Q is $m \times n$ with orthonormal columns, and R is $n \times n$ upper triangular with positive diagonal entries.

Algorithm 2 Gram-Schmidt Orthogonalization

```

1: procedure GRAM-SCHMIDT( $A$ )
2:   for  $j = 1$  to  $n$  do
3:      $\mathbf{v}_j \leftarrow \mathbf{a}_j$ 
4:     for  $i = 1$  to  $j - 1$  do
5:        $r_{ij} \leftarrow \langle \mathbf{q}_i, \mathbf{a}_j \rangle$ 
6:        $\mathbf{v}_j \leftarrow \mathbf{v}_j - r_{ij}\mathbf{q}_i$ 
7:     end for
8:      $r_{jj} \leftarrow \|\mathbf{v}_j\|$ 
9:      $\mathbf{q}_j \leftarrow \mathbf{v}_j / r_{jj}$ 
10:  end for
11:  return  $Q = [\mathbf{q}_1 \cdots \mathbf{q}_n]$ ,  $R = [r_{ij}]$ 
12: end procedure
```

7.4 Singular Value Decomposition (SVD)

Theorem 7.4 (SVD). *Every $m \times n$ matrix A can be factored as:*

$$A = U\Sigma V^\top$$

where:

- U is $m \times m$ orthogonal (columns: left singular vectors)
- V is $n \times n$ orthogonal (columns: right singular vectors)
- Σ is $m \times n$ diagonal with $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$ (singular values)

Proof. Consider $A^\top A$, which is symmetric positive semidefinite. Its eigenvalues are σ_i^2 with eigenvectors $\mathbf{v}_i$. Let $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ for $\sigma_i > 0$. Complete to orthonormal bases. $\square$

Theorem 7.5 (Properties of SVD). 1. $\text{rank}(A) = \text{number of nonzero singular values}$

2. $\|A\|_2 = \sigma_1$ (spectral norm)
3. $\|A\|_F = \sqrt{\sum \sigma_i^2}$ (Frobenius norm)
4. $A = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ (dyadic form)

7.5 Applications of SVD

Theorem 7.6 (Low-Rank Approximation (Eckart-Young)). *For $k < \text{rank}(A)$, the best rank- k approximation in Frobenius norm is:*

$$A_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$$

with error $\|A - A_k\|_F = \sqrt{\sum_{i=k+1}^r \sigma_i^2}$.

Theorem 7.7 (Pseudoinverse via SVD). *If $A = U\Sigma V^\top$, then the Moore-Penrose pseudoinverse is:*

$$A^+ = V\Sigma^+U^\top$$

where Σ^+ replaces nonzero σ_i with $1/\sigma_i$.

8 Functional Analysis

8.1 Metric Spaces

Definition 8.1 (Metric Space). A **metric space** is a set X with a function $d : X \times X \rightarrow [0, \infty)$ satisfying:

1. $d(x, y) = 0$ iff $x = y$
2. $d(x, y) = d(y, x)$ (symmetry)
3. $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality)

8.2 Normed Spaces

Definition 8.2 (Normed Space). A **normed space** is a vector space V with a function $\|\cdot\| : V \rightarrow [0, \infty)$ satisfying:

1. $\|\mathbf{v}\| = 0$ iff $\mathbf{v} = \mathbf{0}$
2. $\|\alpha\mathbf{v}\| = |\alpha|\|\mathbf{v}\|$
3. $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

8.3 Banach Spaces

Definition 8.3 (Banach Space). A **Banach space** is a complete normed space (every Cauchy sequence converges).

8.4 Hilbert Spaces

Definition 8.4 (Hilbert Space). A **Hilbert space** is a complete inner product space.

Example 8.5 (Important Hilbert Spaces). 1. $\ell^2 = \{(x_1, x_2, \dots) : \sum_{i=1}^{\infty} |x_i|^2 < \infty\}$ with $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{\infty} x_i \overline{y_i}$

2. $L^2[a, b] = \{f : [a, b] \rightarrow \mathbb{C} : \int_a^b |f(x)|^2 dx < \infty\}$ with $\langle f, g \rangle = \int_a^b f(x) \overline{g(x)} dx$

8.5 Orthonormal Bases in Hilbert Spaces

Definition 8.6 (Complete Orthonormal System). An orthonormal system $\{e_n\}$ in a Hilbert space H is **complete** if its span is dense in H .

Theorem 8.7 (Parseval's Identity). If $\{e_n\}$ is a complete orthonormal system in H , then for any $f \in H$:

$$\|f\|^2 = \sum_{n=1}^{\infty} |\langle f, e_n \rangle|^2$$

and

$$f = \sum_{n=1}^{\infty} \langle f, e_n \rangle e_n$$

with convergence in norm.

8.6 Fourier Series

Theorem 8.8 (Fourier Series). In $L^2[-\pi, \pi]$, the functions:

$$\frac{1}{\sqrt{2\pi}}, \frac{\cos x}{\sqrt{\pi}}, \frac{\sin x}{\sqrt{\pi}}, \frac{\cos 2x}{\sqrt{\pi}}, \frac{\sin 2x}{\sqrt{\pi}}, \dots$$

form a complete orthonormal system. Any $f \in L^2[-\pi, \pi]$ can be written as:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]$$

where:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$$

9 Linear Operators

9.1 Bounded Linear Operators

Definition 9.1 (Bounded Linear Operator). A linear operator $T : X \rightarrow Y$ between normed spaces is **bounded** if:

$$\|T\| = \sup_{\|x\|=1} \|Tx\| < \infty$$

Theorem 9.2 (Equivalent Characterizations). *For linear operator T , the following are equivalent:*

1. T is bounded
2. T is continuous
3. T is continuous at 0

9.2 Spectrum of an Operator

Definition 9.3 (Spectrum). *For a bounded linear operator T on a Banach space X , the **spectrum** $\sigma(T)$ consists of all $\lambda \in \mathbb{C}$ such that $T - \lambda I$ is not invertible.*

Theorem 9.4 (Properties of Spectrum). 1. $\sigma(T)$ is compact and nonempty

2. $\sigma(T) \subset \{\lambda : |\lambda| \leq \|T\|\}$
3. For self-adjoint operators on Hilbert space, $\sigma(T) \subset \mathbb{R}$

10 Applications to Data Science

10.1 Principal Component Analysis (PCA)

Given data matrix $X \in \mathbb{R}^{n \times p}$ with zero mean columns:

1. Compute covariance matrix $C = \frac{1}{n} X^\top X$
2. Find SVD: $C = V \Lambda V^\top$
3. Principal components are columns of V
4. Projection: $Z = X V_k$ where V_k contains first k eigenvectors

Theorem 10.1 (Variance Explained). *The variance explained by the i th principal component is λ_i , and the total variance is $\sum \lambda_i$.*

10.2 Linear Regression

Given data (X, y) , ordinary least squares solves:

$$\min_{\beta} \|y - X\beta\|^2$$

Theorem 10.2 (Normal Equations). *The solution satisfies:*

$$X^\top X \beta = X^\top y$$

If $X^\top X$ is invertible: $\hat{\beta} = (X^\top X)^{-1} X^\top y$

Theorem 10.3 (Geometric Interpretation). *The least squares solution minimizes the orthogonal distance to the column space of X . The projection matrix is $P = X(X^\top X)^{-1} X^\top$.*

10.3 Regularization

- Ridge regression: $\min_{\beta} \|y - X\beta\|^2 + \lambda\|\beta\|^2$

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top y$$

- Lasso: $\min_{\beta} \|y - X\beta\|^2 + \lambda\|\beta\|_1$

11 Numerical Linear Algebra

11.1 Condition Number

Definition 11.1 (Condition Number). *For a matrix A , the condition number is:*

$$\kappa(A) = \|A\| \|A^{-1}\|$$

For 2-norm: $\kappa_2(A) = \sigma_{\max}/\sigma_{\min}$

Theorem 11.2 (Error Bounds). *For $A\mathbf{x} = \mathbf{b}$ with perturbation δA , $\delta\mathbf{b}$:*

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{\kappa(A)}{1 - \kappa(A) \frac{\|\delta A\|}{\|A\|}} \left(\frac{\|\delta A\|}{\|A\|} + \frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|} \right)$$

11.2 Stability of Algorithms

Definition 11.3 (Backward Stability). *An algorithm for $y = f(x)$ is **backward stable** if the computed $\tilde{y}$ satisfies $\tilde{y} = f(x + \delta x)$ where $\|\delta x\|/\|x\|$ is small.*

12 Advanced Topics

12.1 Tensor Products

Definition 12.1 (Tensor Product). *For vectors $\mathbf{u} \in \mathbb{R}^m$, $\mathbf{v} \in \mathbb{R}^n$, the tensor product $\mathbf{u} \otimes \mathbf{v} \in \mathbb{R}^{m \times n}$ is:*

$$(\mathbf{u} \otimes \mathbf{v})_{ij} = u_i v_j$$

12.2 Kronecker Product

Definition 12.2 (Kronecker Product). *For matrices $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{p \times q}$:*

$$A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{bmatrix} \in \mathbb{R}^{mp \times nq}$$

12.3 Matrix Calculus

Theorem 12.3 (Derivatives). • $\frac{\partial}{\partial \mathbf{x}}(\mathbf{a}^\top \mathbf{x}) = \mathbf{a}$

- $\frac{\partial}{\partial \mathbf{x}}(\mathbf{x}^\top A \mathbf{x}) = (A + A^\top) \mathbf{x}$
- $\frac{\partial}{\partial X} \text{tr}(AX) = A^\top$
- $\frac{\partial}{\partial X} \log \det(X) = X^{-\top}$

13 Exercises and Problems

13.1 Basic Problems

1. Prove that if A and B are invertible, then $(AB)^{-1} = B^{-1}A^{-1}$.
2. Show that $\det(A^{-1}) = 1/\det(A)$ for invertible A .
3. Prove the Cauchy-Schwarz inequality in $\mathbb{R}^n$.
4. Find eigenvalues and eigenvectors of $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

13.2 Intermediate Problems

1. Prove that similar matrices have the same eigenvalues.
2. Show that the SVD of a matrix is essentially unique.
3. Prove Parseval's identity for Fourier series.
4. Derive the normal equations for linear regression.

13.3 Advanced Problems

1. Prove the spectral theorem for symmetric matrices.
2. Show that every matrix has an SVD.
3. Prove the Eckart-Young theorem for low-rank approximation.
4. Derive the Cholesky decomposition algorithm.

Appendix: Useful Formulas

Matrix Identities

$$\begin{aligned}(AB)^{-1} &= B^{-1}A^{-1} \\ (A^\top)^{-1} &= (A^{-1})^\top \\ \det(AB) &= \det(A)\det(B) \\ \text{tr}(AB) &= \text{tr}(BA) \\ \text{rank}(A) &\leq \min(m, n) \text{ for } A \in \mathbb{R}^{m \times n}\end{aligned}$$

Eigenvalue Properties

$$\begin{aligned}\sum \lambda_i &= \text{tr}(A) \\ \prod \lambda_i &= \det(A) \\ \text{For symmetric } A : \lambda_{\min} &\leq \frac{\mathbf{x}^\top A \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} \leq \lambda_{\max}\end{aligned}$$

Norms and Inequalities

$$\begin{aligned}\|\mathbf{x}\|_2 &= \sqrt{\sum x_i^2} \\ \|\mathbf{x}\|_1 &= \sum |x_i| \\ \|\mathbf{x}\|_\infty &= \max |x_i| \\ |\mathbf{a}^\top \mathbf{b}| &\leq \|\mathbf{a}\|_2 \|\mathbf{b}\|_2 \quad (\text{Cauchy-Schwarz}) \\ \|\mathbf{x} + \mathbf{y}\| &\leq \|\mathbf{x}\| + \|\mathbf{y}\| \quad (\text{Triangle})\end{aligned}$$

Fourier Transform Pairs

$$\begin{aligned}f(x) = e^{-a|x|} &\Rightarrow \hat{f}(\omega) = \frac{2a}{a^2 + \omega^2} \\ f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/(2\sigma^2)} &\Rightarrow \hat{f}(\omega) = e^{-\sigma^2\omega^2/2} \\ f(x) = \text{rect}(x) &\Rightarrow \hat{f}(\omega) = \text{sinc}(\omega/2)\end{aligned}$$

References